AI and Machine Learning Workshop Project Report:

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Abstract

The present study had the objective of building an ML to classify dementia stages based on MRI images. Facilitating earlier diagnosis through the detection of subtle disease-related patterns. The three stages of dementia (NoImpairment, MildImpairment, and ModerateImpairment) were classified using a supervised deep learning approach. Specifically, transfer learning technique was employed by using ResNet50V2-based architecture, highlighting the implementation of two-phase training strategy (warm-up and the fine-tune) and a class weight function to solve class unbalance problems. Data were split into training (80%), validation (10%), and test (10%) sets.

On the test set, the model achieved 93.70% accuracy. Additionally, precision was approximately 94% for MildImpairment and NoImpairment, and 90% for ModerateImpairment. Sensitivity reached a macro-average of ~93%, indicating a low false-negative rate. F1-score showed high values (~93%), reflecting balanced precision and sensitivity. Macro-average for specificity was 95.8%. Learning curves showed stable convergence with minimal overfitting, and ROC curves analysis demonstrated that the model has high discriminative ability across all classes.

Ultimately, these findings support the use of ML in healthcare to be use as a decision-support tool by clinicians, in the search to treat neurodegenerative disorders.

Introduction

Dementia, or Major Neurocognitive Disorder (MND) as the DSM-5 defines it now, is a broad term referring to the decline in one or more cognitive domains, such as language, memory, and learning. This pathology can result from various underlying conditions, such as Alzheimer's disease, which is the most common relationship but not the only one, as it can be as extensive as caused even by HIV infection. It is important to clarify that even though the demographic prevalence of this pathology is in older adults, it is not exclusive to this age range. Globally, it affects approximately 47 million people, each of whom develops an assistive dependence and an economic burden for both the family and the public system. In England, the estimated total cost for this illness is around £24.2 billion, with an average cost of £46,050 per individual per year for severe dementia.(1, 2)

To reduce this economic burden on the healthcare system, early diagnosis plays a key role in delaying the progressive symptoms of dementia, allowing therapeutic intervention to be achieved on time. Therefore, in the clinic, Magnetic Resonance Imaging has been widely used to detect structural abnormalities in brain tissue that may indicate neurodegeneration, including regional atrophy, especially within the hippocampus and temporal lobes, as well as alterations in grey and white matter integrity.(2-4)

However, the limitations of these approaches are in their ability to detect unseen patterns beyond human criteria. Machine Learning (ML) can detect subtle patterns that are useful for early detection, staging, and predicting disease trajectory. For instance, it has been reported that an ML approach can differentiate Alzheimer's dementia from Vascular dementia, only employing multimodal MRI by analysing structural white-matter integrity and functional brain connectivity patterns.(4)

Furthermore, for early dementia diagnosis and to aid clinicians with the growing workload, it is necessary to accurately identify the stage of the dementia to arrange cases of urgency. An ML-based model could assist in identifying and classifying based on MRI-derived features.(1, 5)

This report presents a methodological pipeline for the development and evaluation of an ML model to classify dementia MRI images based on their pathological status. The model had the main objective (Table 1) of predicting disease status with performance above chance (e.g., above 70% accuracy), while critically evaluating its design. Finally, the report discusses the ethical aspects of applying ML in the healthcare system, including the benefits and risks of these technologies.

Table 1. Summary of workshop report objectives.

Workshop Report objectives		
Technical objective	**Performance objective**	**Ethical objective**
Develop an ML model for dementia stage classification.	Achieve accuracy above chance (E.g.: 70%), Evaluate model parameters to improve it, and compare results with recent literature.	Analyse benefits and risks of using AI in healthcare.

Methods

Experiment Design

Following the methodological pipeline (Fig.1), the dataset comprised three MRI classes corresponding to dementia stages: NoImpairment, MildImpairment and ModeratImpairment.

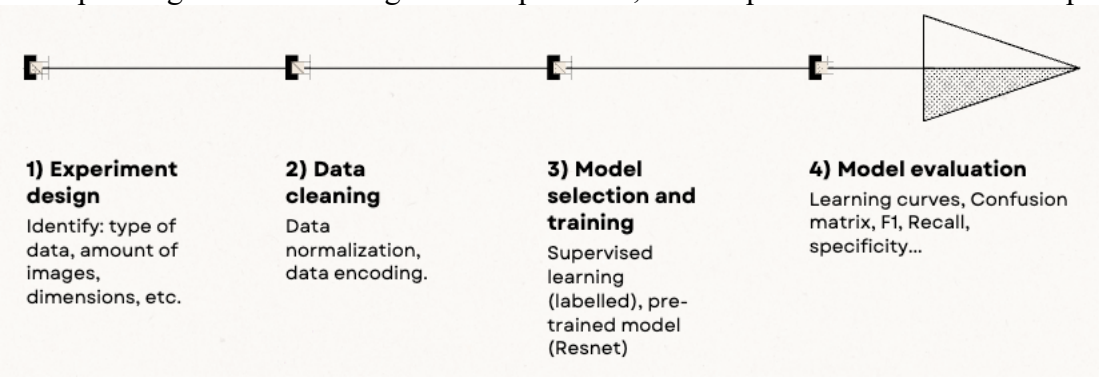

Figure 1. Methodological pipeline. 1) Experiment design, 2) Data cleaning, 3) Model selection and training and 4) Model evaluation.

The dataset exhibits class imbalance across the three categories (Fig.2). All images are encoded in uint8 format, with dimensions of 208x176 pixels and three colour channels.

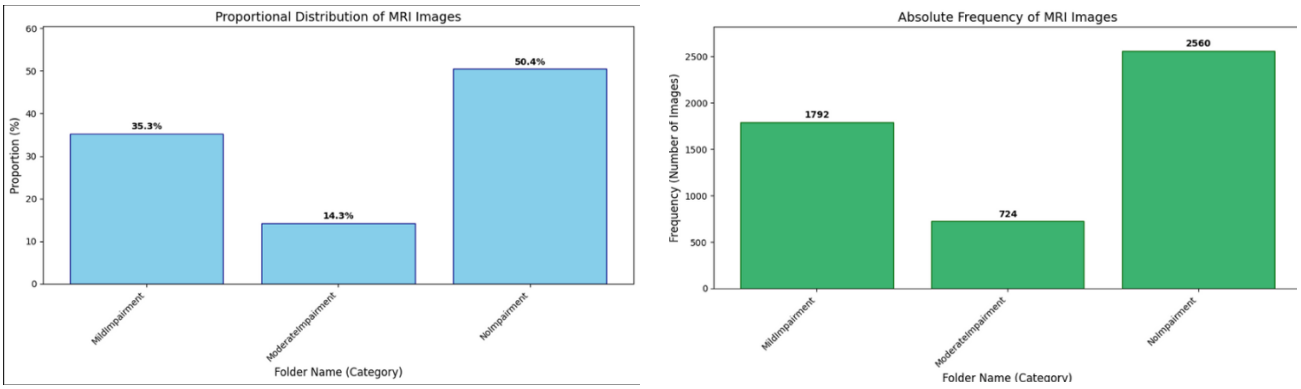

Figure 2. Left: Proportional distribution of MRI images. Right: Absolute frequency of MRI images. Categories: MildImpairment, ModeratImpairment and NoImpairment. The category with a higher number of images is NoImpairment (2560: 50.4%), followed by MildImpairment (1792;35.3%) and ModerateImpairment (724;14.3%). This fact is extremely relevant for training design, as it is necessary to adjust class imbalance.

Data Cleaning

The 3 classes were encoded into class 0 (MildImpairment), 1 (ModerateImpairment) and 2 (NoImpairment). Additionally, all images were normalized to data type float32, dimensions resized into 224x224 and 3 channels RGB.

Data Preprocessing

Data were split (Fig.3) into training (80%), validation (10%) for model tuning and overfitting assessment, and test sets (10%) for evaluation on unseen data.

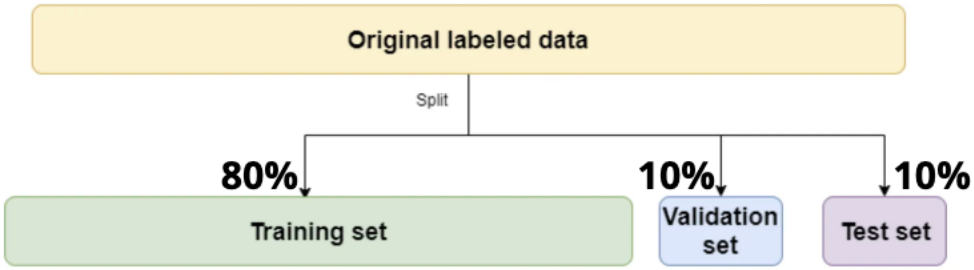

Figure 3. Data splitting. Training set (80%), Validation set (10%), and Test set (10%). This is the validation strategy utilized in this project.

Model Selection

A supervised learning design was chosen as the given data was labelled, and specifically a transfer learning strategy was adopted by using pre-trained ResNet50V2 (Fig.4). This deep convolutional neural network (CNN) has proven ideal for medical images. Particularly, its residual connections are useful for hierarchical feature learning. Additionally, as it is pre-trained in ImageNet improves convergence and generalization.(6)

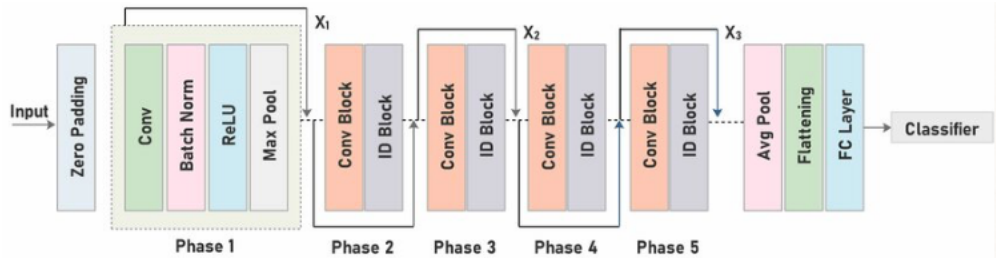

Figure 4. RestNet50v2 architecture. It is made up of 50 trainable layers and 16 residuals blocks. (7)

Model Architecture

The custom model architecture design used ResNet50V2 backbone with the original classifier removed. A preprocess layer was added to ensure input quality. Data augmentation layer was applied exclusively to the training set to improve generalisation and reduce overfitting, with RandomZoom, RandomBrightness, and RandomConstrast. Feature extraction was performed using GlobalAveragePooling2D, followed by the layers Dense (256), BatchNormalization, Dropout (0.3) and Dense (128). And a softmax output layer for three-class classification. (6, 8)

The Dense 256 layer was added to capture more complex patterns. BatchNormalization was added to stabilise training and dropout to reduce overfitting.

Model Training

In model training, Adam optimizer was used along the sparse_categorical_crossentropy as loss function, suitable for multi-class classification with integer-encoded labels. To handle class imbalance, class weights were applied during training, assigning higher penalties when mistakingly assignly the less represented class (MildImpairment), forcing the model to not ignore them. A fixed batch size (32) was used in training. EarlyStopping (8) and ReduceLROnPlateau callbacks were employed to prevent overfitting and improve convergence.

Training used a two-phase training (50 epochs): a warm-up phase (LR=1e-4) with a frozen backbone and a fine-tuning phase (LR= 1e-5) to adapt pretrained features. The final outputs include the trained ".keras" model and two ".json" log files.(9)

Model evaluation

The chosen parameters calculated are showed in Table 2, including a brief description of their purpose.

Table 2. Model evaluation parameters summary.

Training Dynamics	Classification Quality	Decision Performance
Learning Curves(Train vs Val Loss/Accuracy) Detect overfitting & underfitting	**Confusion Matrix** Identify class confusions	**ROC Curves (OvR)** Compare class separability
Generalization Check	**Error Balance**	**Threshold-free Metric**
Validation vs Test metrics Stable gap → good generalization	**Precision • Recall • F1-score** Balance FP vs FN	**AUC** Higher AUC → better model

Results

The classification report (Table 3) and confusion matrix (Fig. 5) for the final model provide evidence for model validation based on the test set (unseen data). Table 3 shows an accuracy of 93.7%, demonstrating that the model correctly classifies most images. This result aligns with the confusion matrix, where the diagonal elements represent correctly predicted cases.

Table 3. Classification report.

	Precision	Sensitivity (recall)	Specificity	F1-score	Support
Class 0 Mild	0.947	0.905	0.973	0.926	179
Class 1 Moderate	0.895	0.932	0.982	0.913	73
Class 2 Noimp	0.943	0.961	0.94	0.952	256
Accuracy	0.937				
Macro avg	0.928	0.932	0.965	0.93	508
Weighted avg	0.937	0.937	0.958	0.937	508

In terms of precision, MildImpairment and NoImpairment yielded similar results around 94%, while ModerateImpairment had a slightly lower value of ~90% (Table 3). This indicates that the false-positive (FP) rate is low, and ModerateImpairment is prone to greater value fluctuations than the other classes, since it has the fewest images.

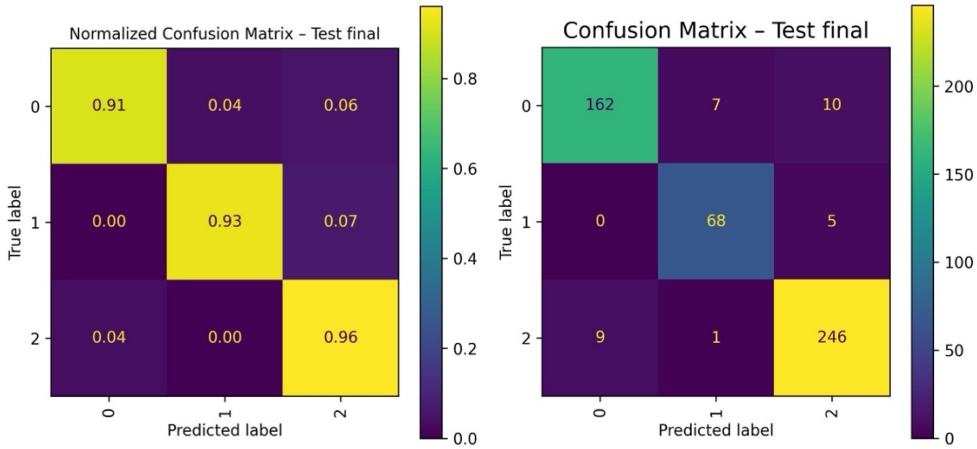

Figure 5. Confusion matrix. Left: normalised. Right: non-normalised. Class 0 = Mild, Class 1 = Moderate and Class 2 = No impairment. Showing high diagonal values across all three classes (0.91-0.96), indicating good classification performance across classes.

Sensitivity shows a macro- and weighted-average of around 93% (Table 3), indicating an overall low false-negative (FN) rate. MildImpairment accounts for 6% of its cases being predicted as NoImpairment, which is one of the most common confusions in the experiment, explaining why the sensitivity in MildImpairment is the lowest. Consequently, the macro and average F1-scores are high (~93%), which is explained by the strong balance between sensitivity and precision.

Specificity values (Table 3) are above 94%, with a macro-average of 96.5%. This indicates that the model effectively minimizes false-positives predictions across all classes.

Learning curves (Fig. 6 and 7) showed stable learning, as evidenced by a steady increase in accuracy and a plateau around 30 epochs. This is further supported by the progressive decrease in loss, which reaches a plateau around 30 epochs and converges there. A small amount of overfitting is evident in both plots, as there is a small gap (~0.08) between training and validation accuracy, and another (~0.2) between training and validation loss.

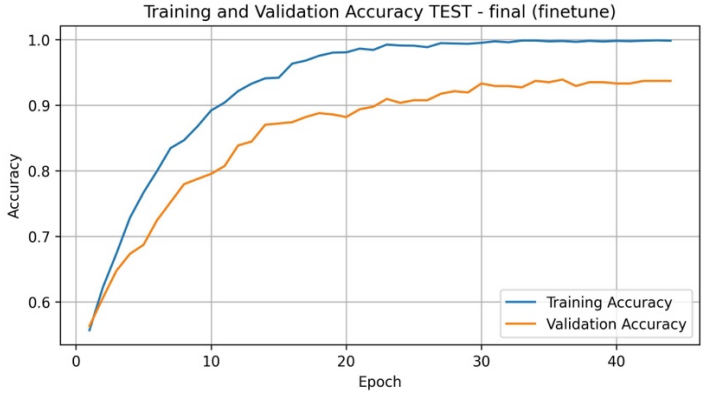

Figure 6. Learning Curves: Training and Validation accuracy across epochs. Training accuracy increased progressively across epoch (expected behaviour). Optimal performance at around 30 epochs. Small overfitting of around 0.08.

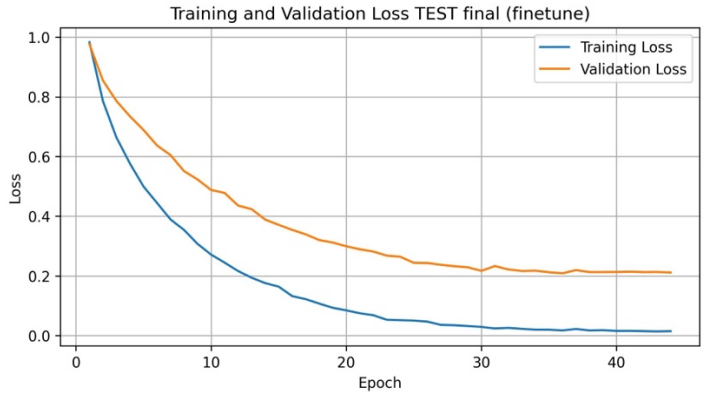

Figure 7. Learning curves: Training and Validation loss across epochs. The training and validation loss decreased progressively during training (expected behaviour). Optimal performance around 30 epochs. Small overfitting of around 0.2.

ROC curves (Fig.8) indicate that the model has strong discriminating ability across decision thresholds, as all the curves are on the top left of the graph, and all the classes scored high AUC values.

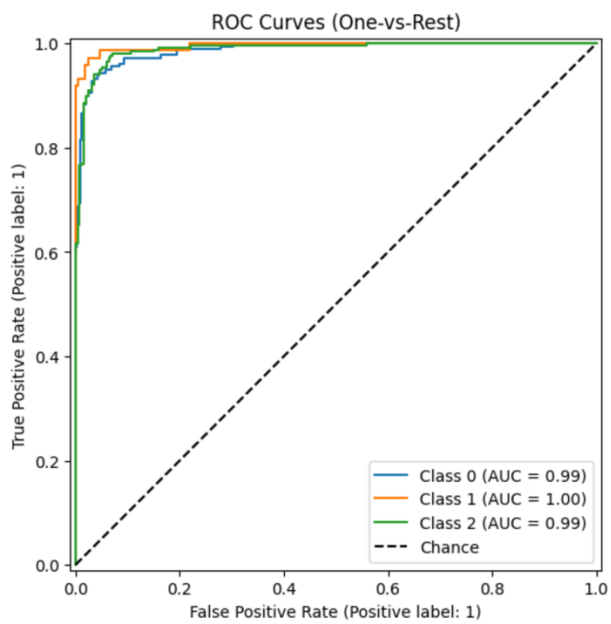

Figure 8. ROC curves. Representing good separability ability from the model. Class 0 AUC = 0.99, Class 1 AUC = 1.00 and Class 2 AUC = 0.99.

Discussion

When developing the model, one of the biggest limitations was how to handle class imbalance: in its early stages, the model had an accuracy of 52.8%, a systematic bias toward overpredicting NoImpairment, and ModerateImpairment (the fewest images) was never predicted. When adding the class-weight function to the training, there was a substantial increase in accuracy and a corresponding reduction in confusion between classes. However, this was not enough to completely solve the problem of ModerateImpairment prediction.

Another limitation was that data augmentation was too harsh on the dataset, so diminishing its severity improved model performance. This is probably because severe data augmentation caused the images to change to a point where they were no longer representative of their own class, and the model was not learning pathological features but instead artefacts. Additionally, the inclusion of Dropout, to avoid overfitting, which randomly deactivates neurons during training and improves generalization.

One of the model's strengths lies in the use of a two-phase training strategy, which is designed to improve model performance, likely because the backbone adapts better to the MRI domain and thereby improves feature specialization. Moreover, although the model showed positive results, applying K-fold cross-validation would provide a more robust estimate of its overall performance and generalisation.

From a clinical perspective, the fact that MildImpairment and NoImpairment are the classes that are most often confused (Fig. 5) can be a problem. Confusing MildImpairment with NoImpairment is a FN in the context of early-stage detection, meaning that an early intervention could be missed, and what is worse, it can silently escalate into Moderate dementia. The reason may be that some pathological features are not fully developed, so the model misses them. What is important is to ensure that the clinician is aware of these cases and investigates them in detail. Additionally, adding Grad-Cam to the model evaluation could help better understand what the model is learning, specifically from the images.(10)

Rao *et al.* (2024) implemented a similar ResNet50V2 model to classify three stages of Alzheimer's dementia using 3D MRI images, achieving 91.25% test accuracy. In comparison, the testing accuracy in this experiment was 93.70%. Although the imaging dimensionality differs, these results suggest that ResNet-based architectures remain effective for MRI-based dementia classification.(11)

Similarly, Stoleru & Iftene (2023) utilized a ResNet152 approach for binary classification of Alzheimer's utilizing MRI images (2D) of multiple slices. They reported a testing accuracy of 99.96%, which is higher than that in this experiment. Although their experiment is not a multi-stage categorization, it is still relevant to consider that a version with more layers of ResNet could help to achieve better testing accuracy results. Lastly, Khan *et al*. (2023) performed a multiclass classification of Alzheimer’s disease MRI images through a VGG19 model, reporting very high accuracy (98.47%) and F1-score. Given that their classification task closely resembles the objectives and framework of the present study, exploring VGG19 in future studies could be valuable to assess whether similar performance improvements can be achieved.(12, 13)

Table 4. Comparative table. Comparison of the present study to literature, highlighting differences in model architecture, input dimensionality and classification framework. (11-13)

Study	Architecture	Dimensionality	Classification task	Test accuracy
Present study	ResNet50V2	2D slices	3-classes	93.70%
Rao *et al.* (2024)	ResNet50V2	3D volumes	3-classes	91.25%
Stoleru & Iftene (2023)	ResNet152	2D slices	Binary	99.96%
Khan *et al.* (2023)	VGG19	2D slices	4-classes	98.47%

Ethical implications

AI has demonstrated the potential to increase diagnostic accuracy, for instance, through analysing hidden pathological patterns in X-rays beyond human ability. Subsequently, these ML models' processing functions can potentially reduce clinician workload and support cost-effective therapy decisions, thereby benefiting large public systems such as the NHS. (14, 15)

However, developers have a responsibility to communicate the potential risks and limitations of their models to clinicians, as excessive reliance on AI may compromise clinical judgment and lead to misdiagnosis. Importantly, in terms of roles, AI should serve as a decision-support tool rather than a medical replacement. For instance, ML models can flag borderline cases for clinicians to review more carefully, particularly when an FN would carry substantial clinical risk.(14, 15)

Moreover, the patient's autonomy principle must be respected, which means informed consent is required. Even before the AI is deployed, there must be a guarantee that their personal sensitive data is secured. Lastly, according to the principle of justice, socio-economic status should not limit access to these technologies, particularly if they are central to enabling a paradigm shift to a precision medicine-based healthcare system. Therefore, it is also important to consider that the dataset used to train the model is as representative as possible, meaning that developers need to make sure not only to cover a significant number of cases related to the disease but also to include different ethnicities, sexes, ages, and types of devices used to gather the data.(14, 16-18)

Table 5. Summary of Benefits and Risks/Challenges of AI in healthcare.

Benefits	**Risk/Challenges**
Detects hidden pathological patterns	Overreliance may compromise clinical judgment
Reduced clinician workload	Risk of misdiagnosis
Cost-effective therapy decisions	AI should be treated as a decision-support tool
Scalability in public systems (e.g., NHS)	Informed consent is required
Selects borderline cases for review	Inequitable access due to socio-economic factors.

Conclusion

In summary, the results obtained from ResNet50V2-based model probed to accomplish the multi-class classification objective for dementia stages. This is reflected in the testing accuracy and F1 scores, the latter probing that the model has a good equilibrium between precision and sensitivity. Moreover, all the classes probed AUC values near 1, highlighting the model separability qualities. Importantly, this study implicates its importance in supporting the development of a future model that could accurately predict dementia stage and consequently facilitating care planning.

However, in the future is necessary to perform a K-fold validation test in order to assure overall robustness of the model and applying tools as Grad-Cam to further evaluate which patterns is the model learning from.

Ultimately, AI in healthcare relies not only in its technical performance but in the responsibility of developers on working alongside clinicians to create decision-support tools that would improve people life, while upholding principles of autonomy, beneficence, non-maleficence and justice.

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